3.2.1 Refraction and Snell's law

Light changes speed between media. At oblique incidence it bends towards the normal on entering a slower, higher-index medium, and away on entering a faster one. Snell's law relates the angles, both measured from the normal. At normal incidence the speed can change without a change in direction. Frequency remains unchanged across the boundary.

$$\frac{\sin i}{\sin r} = \text{constant}$$

$$n_X = \frac{\sin\theta_a}{\sin\theta_X}$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Incident ray, refracted ray and normal lie in one plane. i, r are measured from the normal; fixed pair of media and fixed wavelength.

n_X: refractive index of medium X relative to vacuum (air approximated as n = 1). Angles in air and X are measured from the normal.

Snell's law: n_1, n_2 are refractive indices; θ_1, θ_2 are angles from the normal, not the surface.

Refractive index and light speed

Refractive index is vacuum light speed divided by light speed in the medium at the specified frequency. A higher index therefore means slower light. Index is dimensionless and can depend on wavelength. Across a boundary, frequency remains constant while wavelength changes with speed. Air is normally approximated as having index 1.

$$n = \frac{c}{v}$$

n: refractive index; c: light speed in vacuum, approximately 3.00×10^8 m s^{-1}; v: light speed in the medium at that frequency.

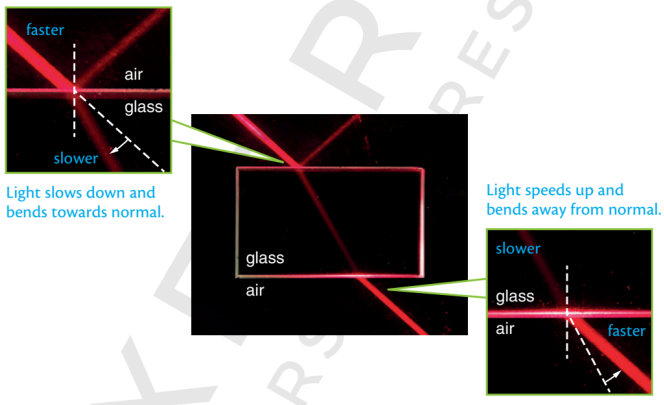
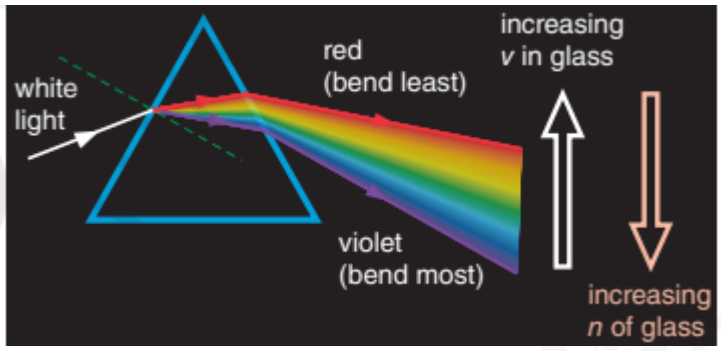


Fig 2.1e Light ray passing through a glass block.

At the air-to-glass boundary light slows and bends towards the normal; on leaving glass it speeds up and bends away. The frequency stays unchanged.

3.2.2 Dispersion

Different colours can travel at different speeds and have different refractive indices in the same medium. A prism then deflects them through different angles, separating white light into colours: dispersion. In ordinary glass over the visible range, violet is deviated more than red. Crossing the boundary does not change each colour's frequency.



Different colours have different refractive indices in the prism and are deflected through different angles.

3.2.3 Critical angle and total internal reflection

For light travelling from higher to lower refractive index, increasing incidence angle brings the refracted ray closer to the boundary. The critical angle gives a refracted angle of 90°. Total internal reflection occurs only above this critical angle. At equality, the refracted ray runs along the boundary. All angles are measured from the normal.

$$\sin C = \frac{1}{n}$$

C: critical angle measured from the normal; n: refractive index of the denser medium. This form assumes the other medium is air/vacuum; generally $\sin C = n_2/n_1$ with $n_1 > n_2$.

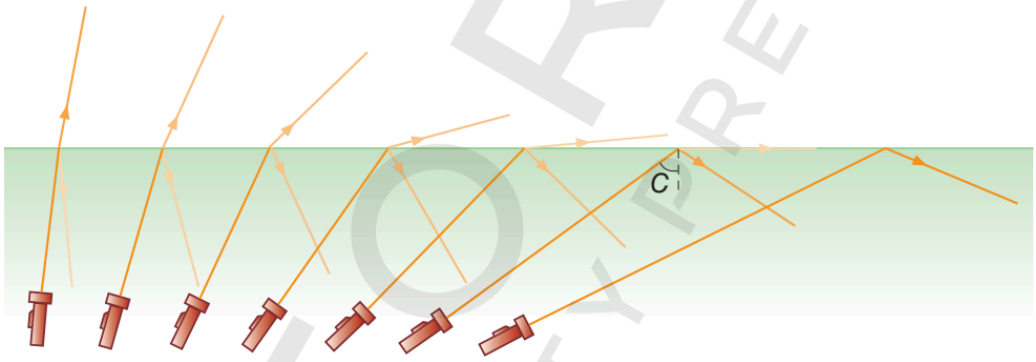


Fig 2.2b *Total internal reflection occurs only when the angle of incidence is larger than C.*

At the critical angle the refracted ray runs along the boundary. Total internal reflection occurs only for incidence greater than C from the higher-index side.